Matrix inequalities and norms

Concave unitary-orbit subadditivity without monotonicity

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Removing monotonicity from a unitary-orbit comparison is challenging; spectral bounds for broad classes of matrix functions have community importance.

Problem statement

For every integer $n \geq 1$, positive semidefinite matrices $A, B \in \mathbb{C}^{n\times n}$, and real-valued concave function $f : [0, \infty) \to \mathbb{R}$ with $f(0) \geq 0$, must there exist unitary matrices $U, V \in \mathbb{C}^{n\times n}$ such that

$$f(A+B) \preceq Uf(A)U^* + Vf(B)V^*?$$

Here $X \preceq Y$ means $Y - X$ is positive semidefinite, $U^*U = UU^* = I$, and $f(A)$ is defined by applying f to the eigenvalues in a spectral decomposition of A. Concavity means

$$f(\theta x + (1-\theta)y) \geq \theta f(x) + (1-\theta)f(y) \quad (x, y \geq 0,\ 0 \leq \theta \leq 1).$$

The function is not assumed nonnegative on its entire domain; adding that assumption would change the problem. The unitaries may depend on A, B, f.

The requested inequality is an order comparison between sums of matrix functions up to changes of orthonormal basis. Such comparisons yield spectral and norm bounds for nonlinear transformations of positive semidefinite matrices.

References

1. K. M. R. Audenaert and F. Kittaneh, *Problems and Conjectures in Matrix and Operator Inequalities*, arXiv:1201.5232v3 (2012), §2, Problem 5 and equation (11). Full text.
2. J.-C. Bourin and E.-Y. Lee, *Unitary orbits of Hermitian operators with convex or concave functions*, Bull. Lond. Math. Soc. 44 (2012), 1085–1102, Theorem 3.1 and Remark 3.13. Preprint.

Status check (2026-09-10): the two sources explicitly leave removal of monotonicity open. The published 2017 Audenaert–Kittaneh update also retains the question. Searches for “concave unitary subadditivity monotonicity”, “Bourin Lee nonmonotone concave”, and 2025/2026, together with the new arXiv:2609.05854 paper on Horn and orbit inequalities, found no stated resolution of this full function class. Recent results for nonnegative concave functions impose a stronger assumption. This is a bounded status check.